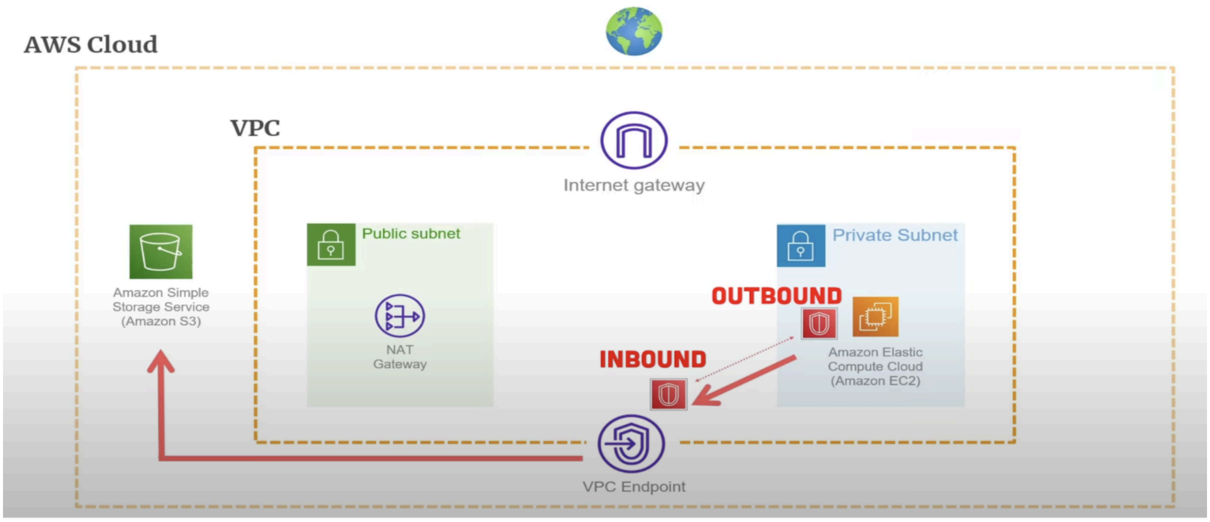
VPC Endpoint ile S3 Bucket'a erişim

Bir şirkette güvenlik gereği **private subnet içindeki EC2 instance**'ların internete çıkması yasak. Ancak bu EC2'lerin **S3 bucket'a (örn. log yedekleri için)** erişmesi gerekiyor.

- Normalde EC2 → S3 için internete çıkış (NAT Gateway/IGW) gerekir.
- Fakat burada **VPC Endpoint (Gateway Endpoint)** kullanarak EC2 → S3 iletişimini AWS'in private network'ü üzerinden sağlayacağız.

👉 Böylece:

- İnternete çıkış olmayacak, güvenlik artacak.
- Trafik AWS backbone üzerinden S3'e gidecek.
- Daha ucuz (NAT Gateway ücreti yok).



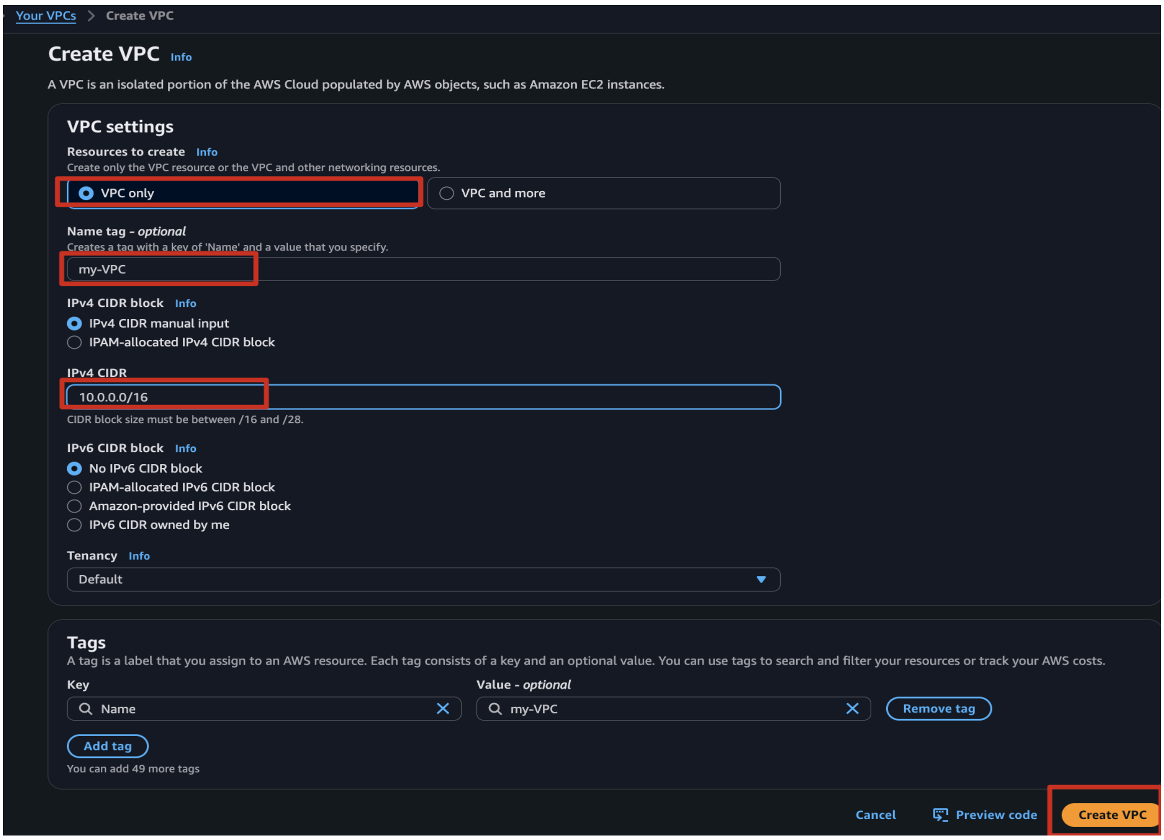
🛠️ Lab Adımları

1️⃣ Ön Hazırlık – VPC & Subnet

1. AWS Console → VPC Dashboard → **Create VPC**
   - VPC CIDR: `10.0.0.0/16`
   - Subnetler:
     - **Public Subnet**: `10.0.1.0/24`
     - **Private Subnet**: `10.0.2.0/24`
   - Public Subnet'e IGW bağla (Bastion Host için kullanılabilir).
   - Private Subnet → Route Table sadece `Local` entry ile olacak (0.0.0.0/0 yok).

VPC ve Subnet'ler

- AWS Console → VPC → "Create VPC"
  - **Vpc Settings:** `VPC only`
  - **Name tag:** `my-VPC`
  - **IPv4 CIDR block**: `10.0.0.0/16`

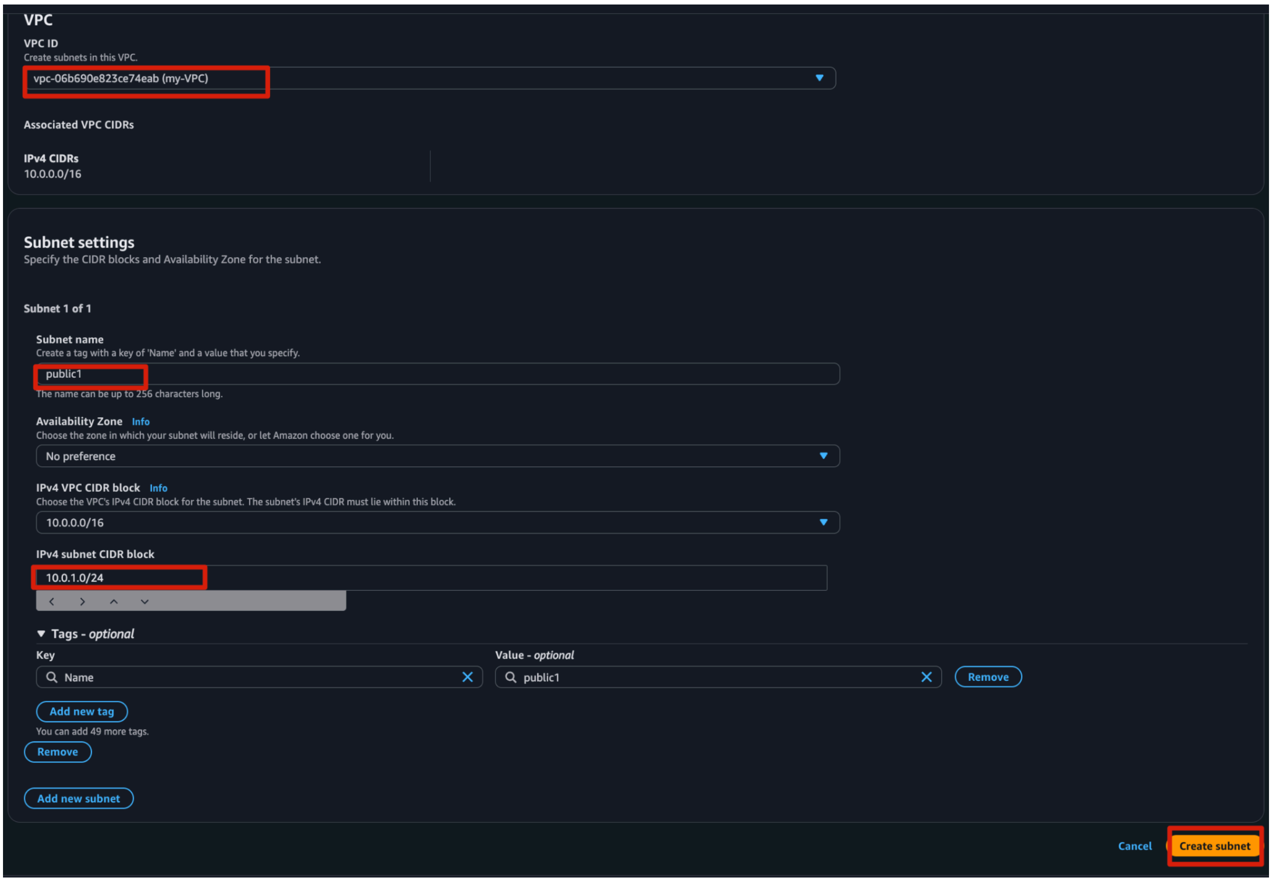


Subnet oluşturma:

- Public Subnet

  Subnets→Create Subnet

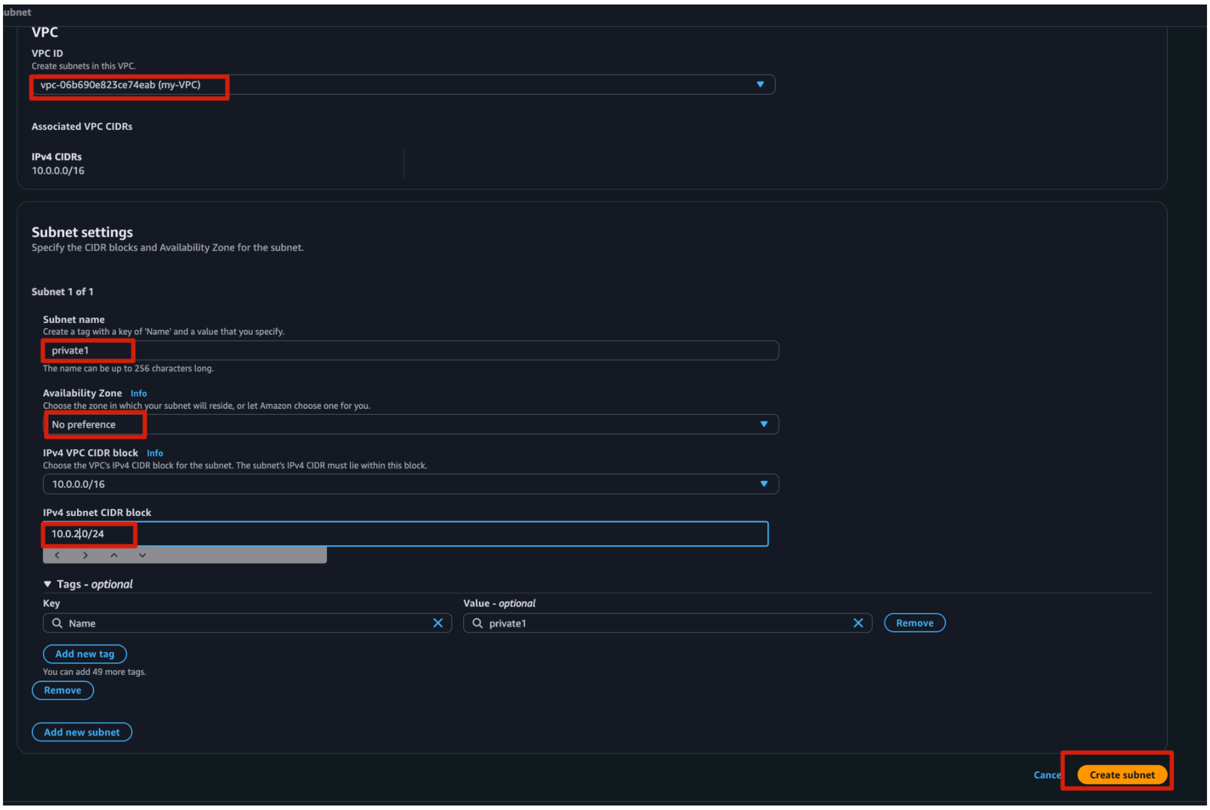
  - **VPC ID**: `my-VPC`
  - **Subnet Name**: `public1`
  - **Availability Zone**: `No preference`
  - **IPv4 subnet CIDR block**: `10.0.1.0/24`



- Private Subnet

  Subnets→Create Subnet

  - **VPC ID:** `my-VPC`
  - **Subnet Name**: `private1`
  - **Availability Zone:** `No preference`
  - **IPv4 subnet CIDR block**: `10.0.2.0/24`

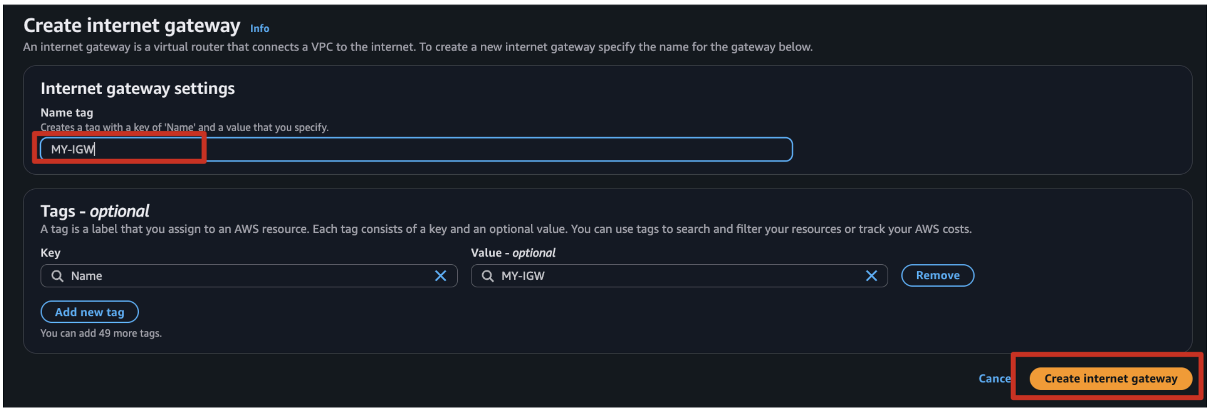


Internet Gateway (IGW)

VPC→Internet gateways→Create internet gateway

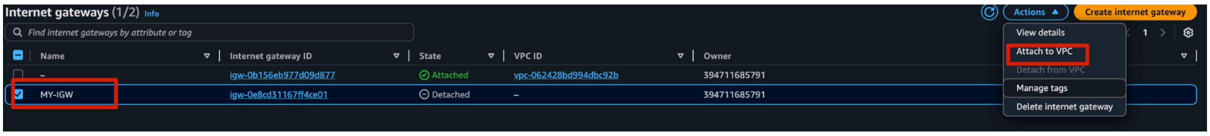
Internet gateway settings:

- Name: `MY-IGW`

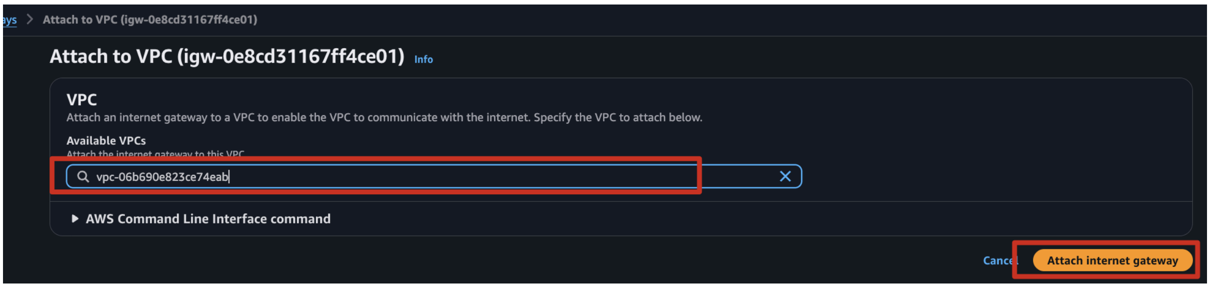


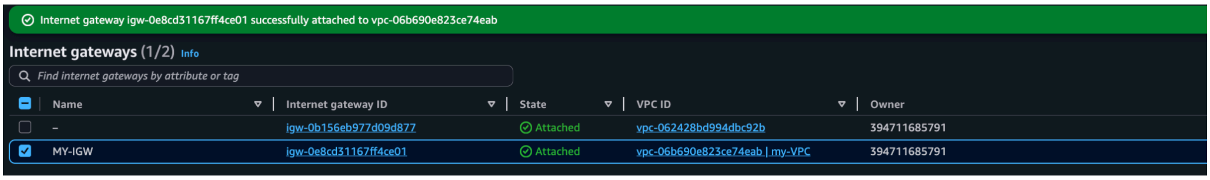
Internet gateway VPC’ye attach etme

internet gateway seç **Actions** dan **Attach to VPC** seç



VPC içerisinden **My-VPC** seç ve **Attach internet gateway** butonuna bas.



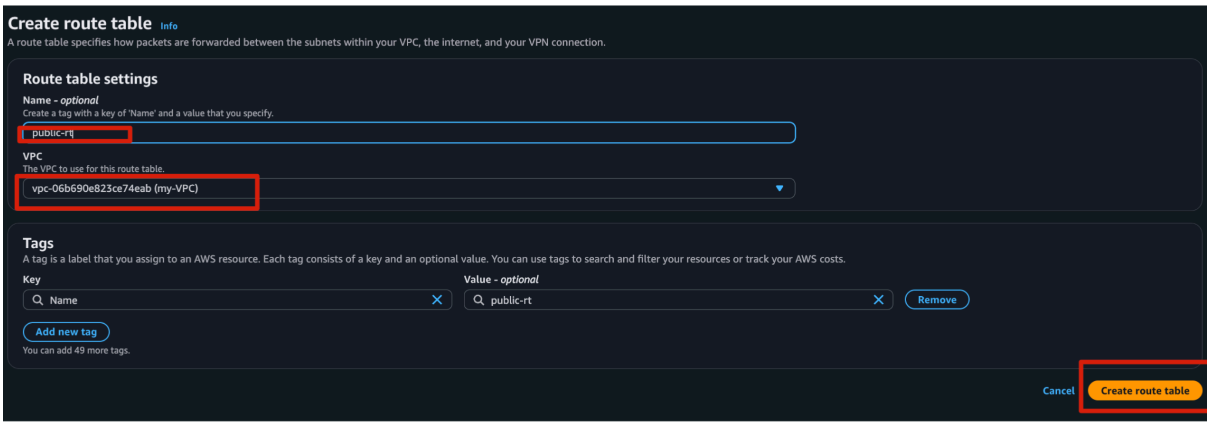


Route Tables

- Public Route Table:

VPC—>Route tables→Create route table

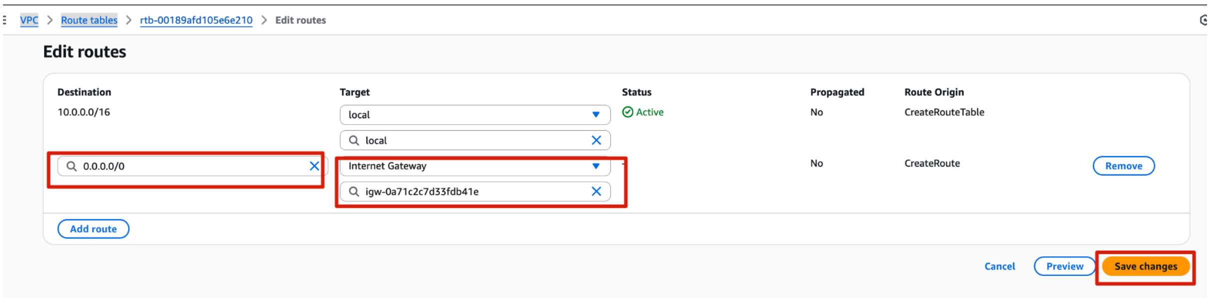
  - Name: `public-rt`
  - `VPC: my-VPC`



Route Ekle

public-rt→Routes→Edit routes–Add route

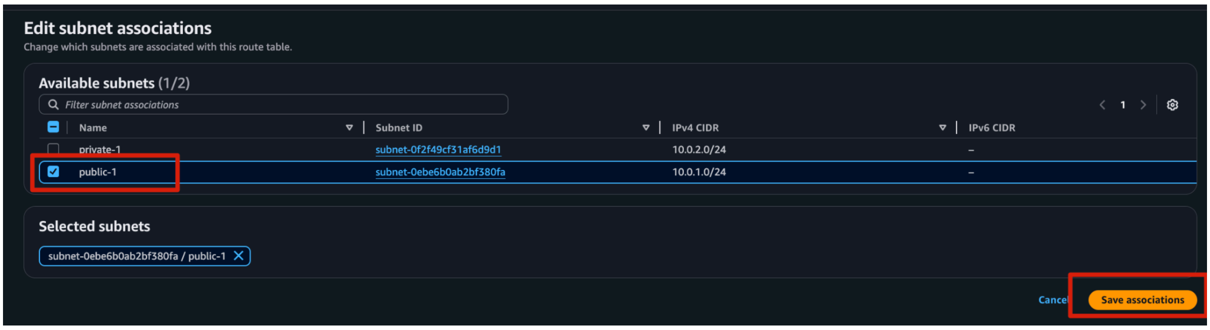
- Destination: `0.0.0.0/0`
- Target: `internet Gateway(MY-IGW)`



Route Table'ı Public Subnet ile ilişkilendir

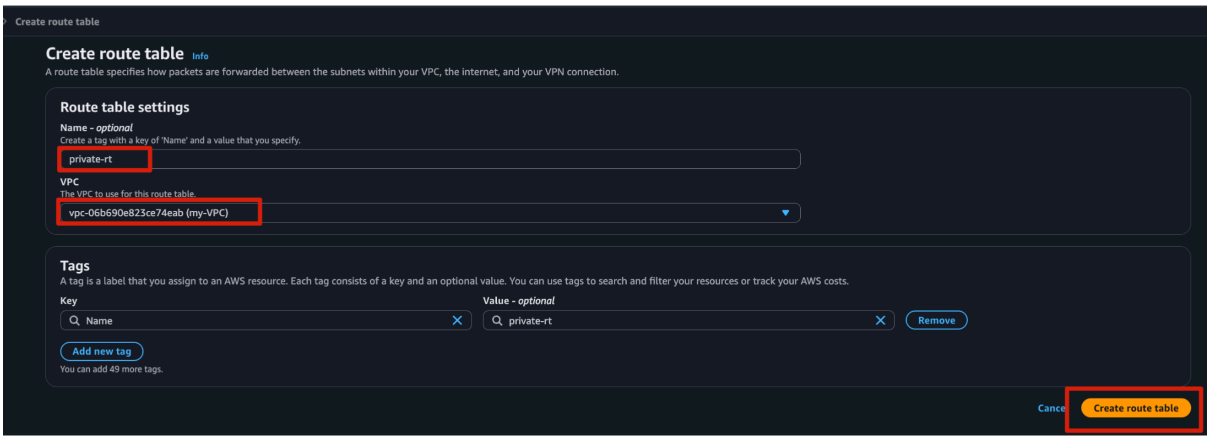
public-rt→Subnet associations→Edit subnet associations→Select Public-1

- **Available subnets:** `public-1`



VPC—>Route tables→Create route table

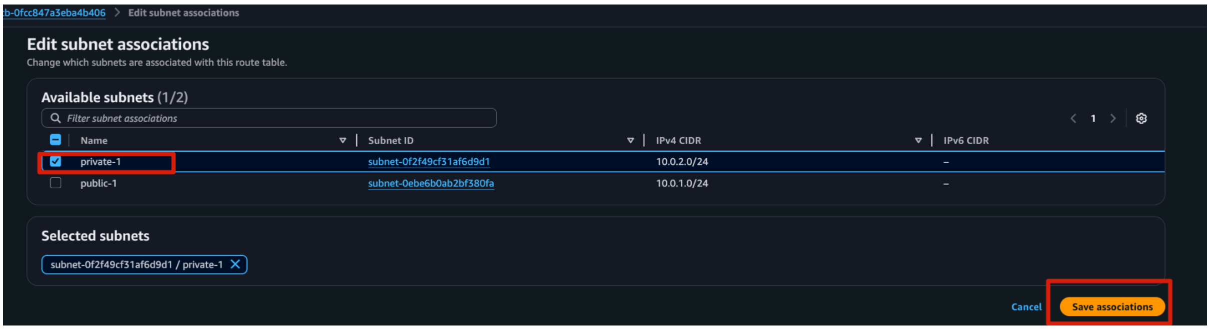
- Name: `private-rt`
- `VPC: my-VPC`



Route Table’ı Private Subnet ile ilişkilendir

private-rt→Subnet associations→Edit subnet associations→Select private-1

- **Available subnets**: `private-1`

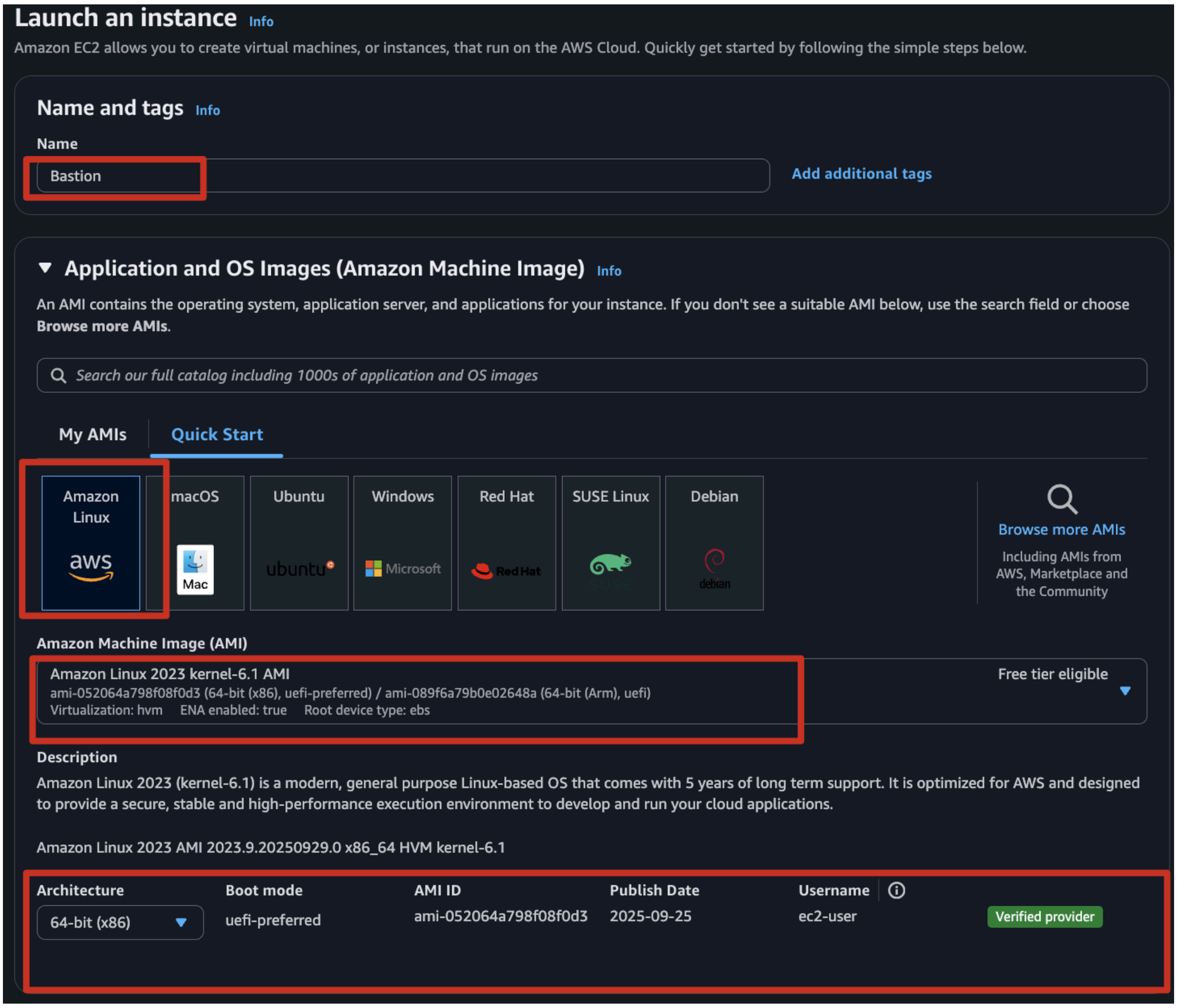


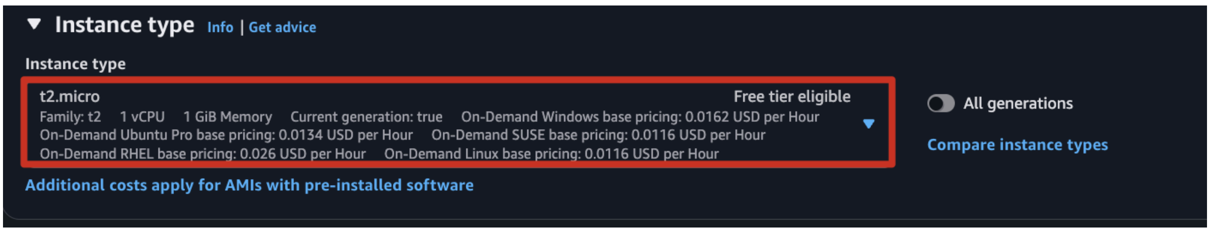
2️⃣ EC2 Instance Kurulumu

1. **Public Subnet’te** bir Bastion EC2 instance oluştur:

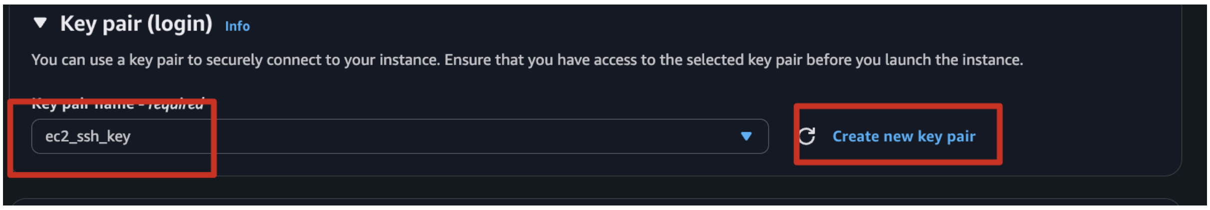
EC2→Instances→Launch an instance

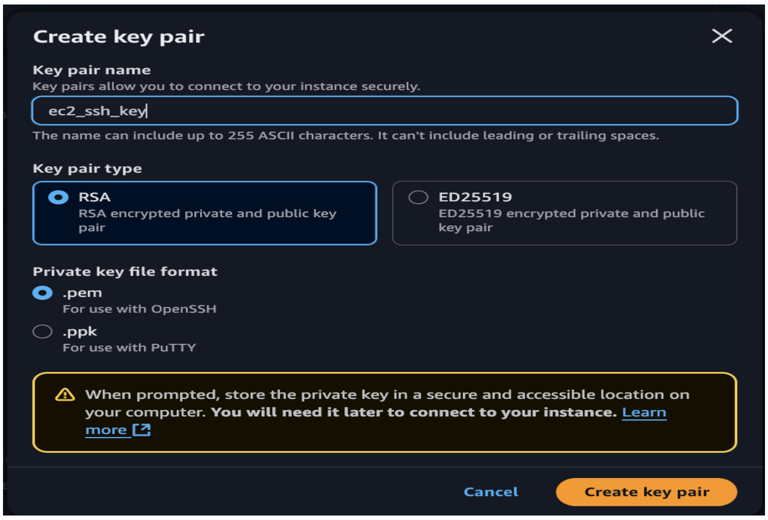
- **Name and tags**: `Bastion`
- **AMI**→ `Amazon Linux 2023, t3.micro`



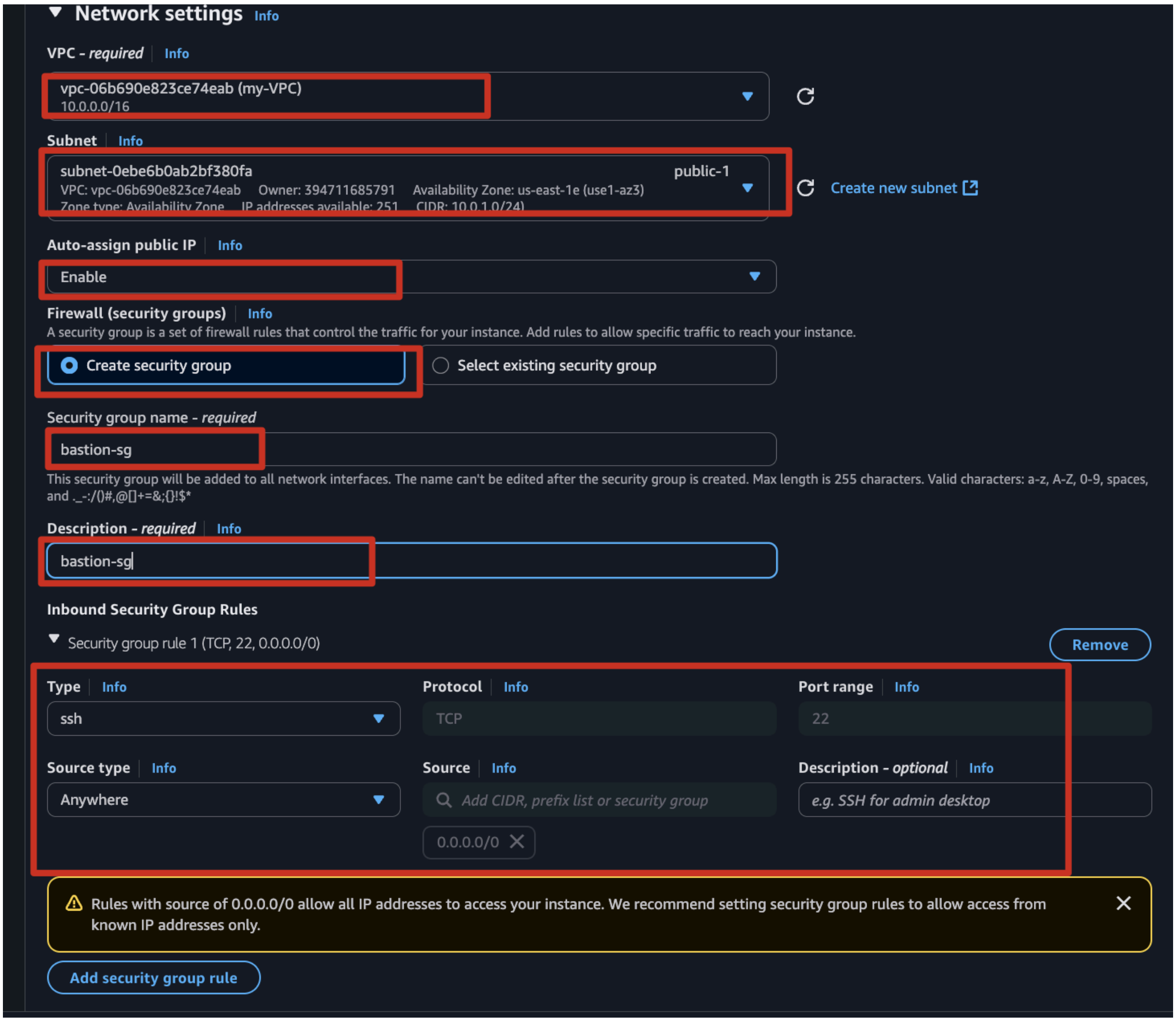


- **Key Pair (login):**
  - **Create new key pair → Key pair name:** `ec2_ssh_key`
  - **Type:** RSA, **File format:** `.pem` → **Create** ve seç.

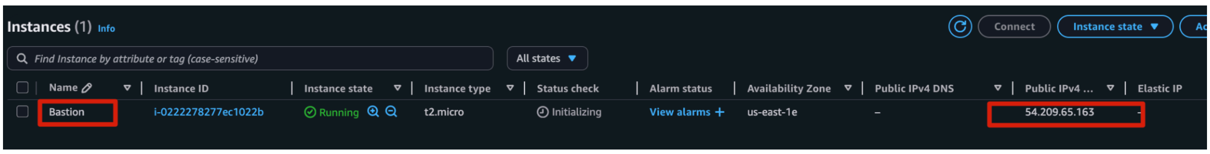




- **Network settings → Edit:**
  - **VPC:** `my-VPC`
  - **Subnet:** `public-1`
  - **Auto-assign public IP:** `Enable`
  - **Firewall (SG): Select security group**
    - **Security group name:** `bastion-sg`
    - **Description:** `bastion-sg`
    - **Inbound rules:**
      - **Type**: `SSH`
      - **Protocol:** `TCP`
      - **Port range:** `22`
      - **Source**:`0.0.0.0/0`



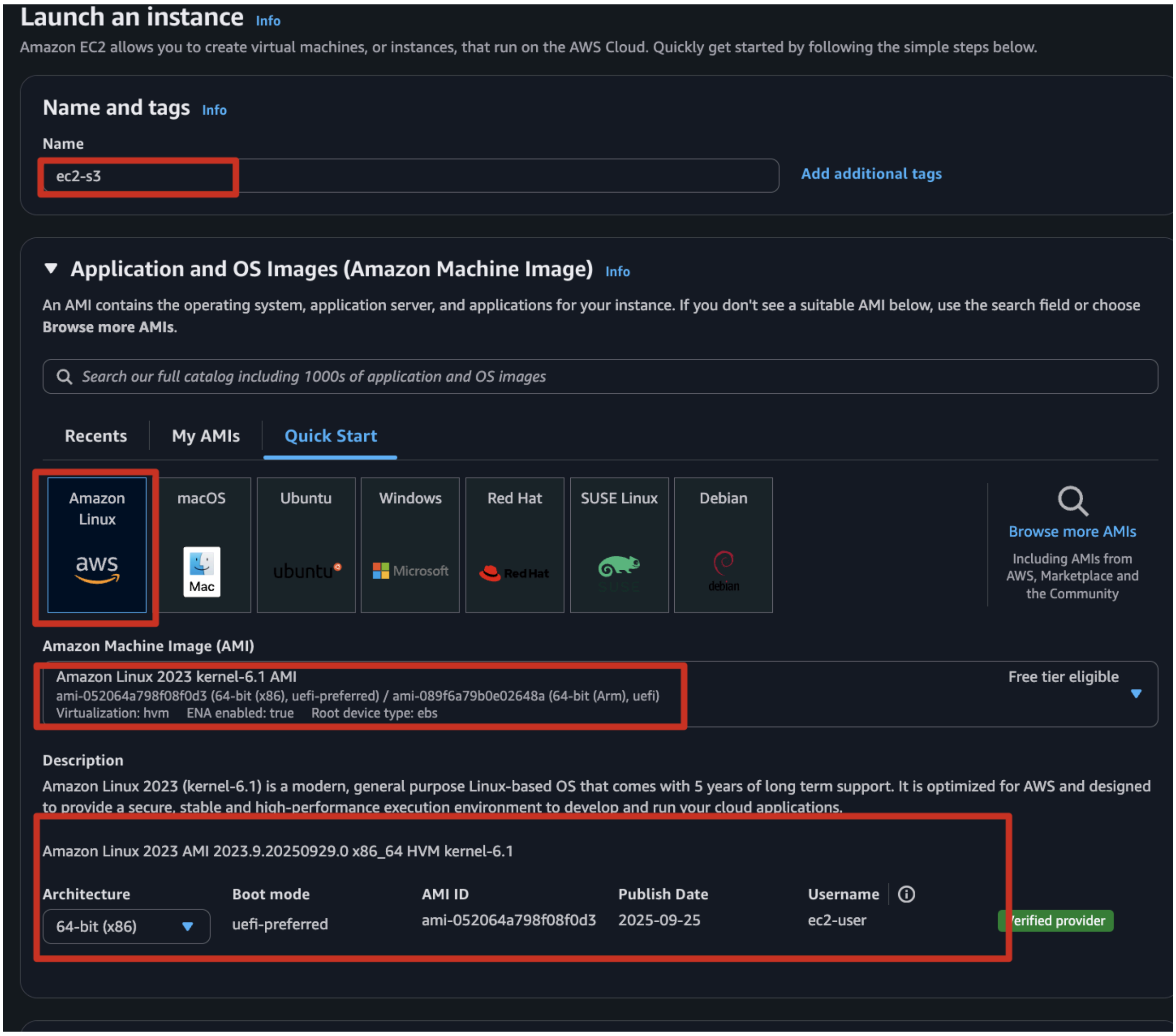
Launch instance butonuna tıklayınız

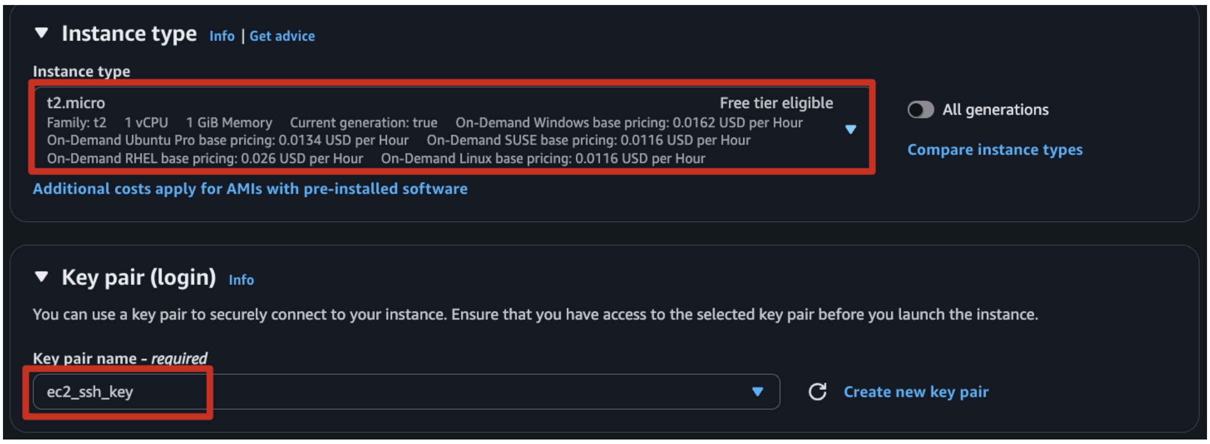


2. **Private Subnet'te** bir EC2 instance oluştur:

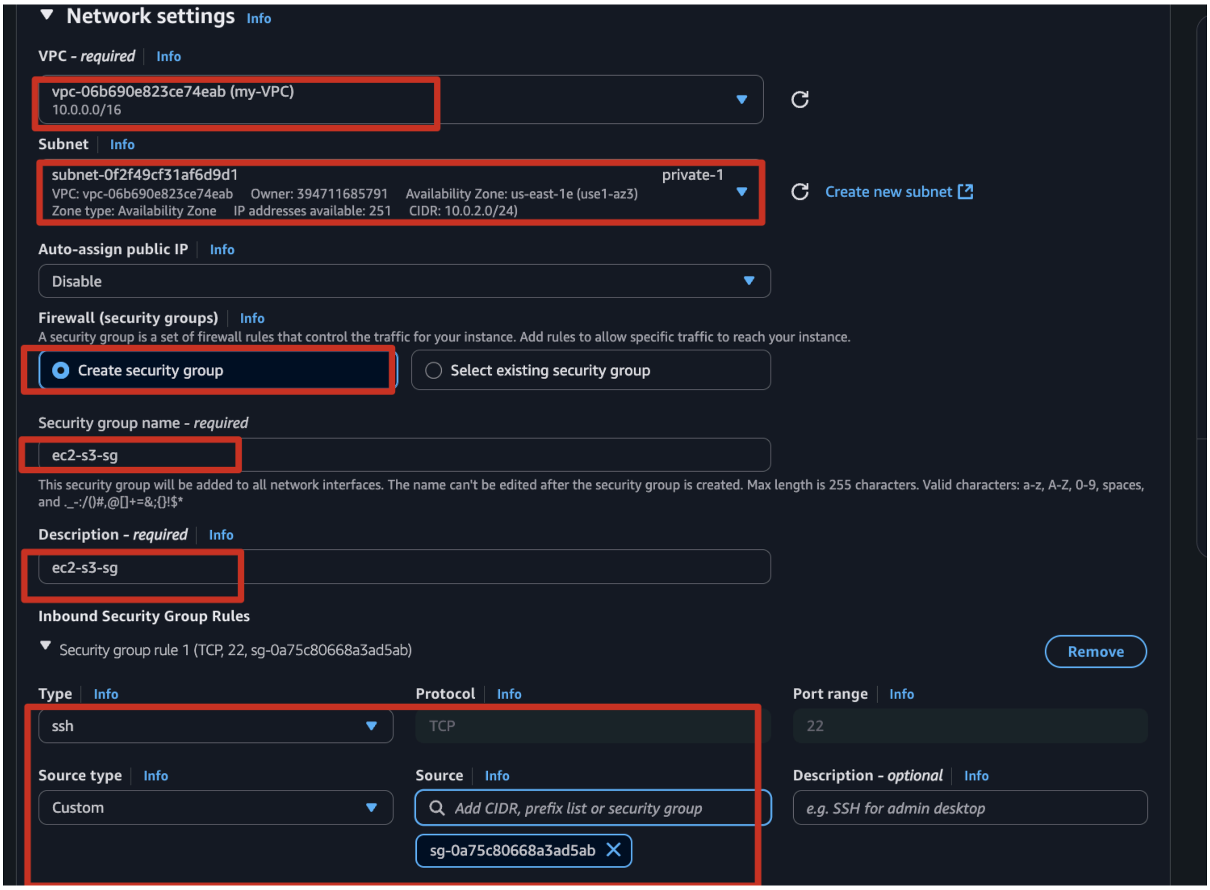
EC2→Instances→Launch an instance

- **Name and tags**: `ec2-s3`
- **AMI**→ `Amazon Linux 2023, t2.micro`
- **Key pair name:** `ec2_ssh_key`

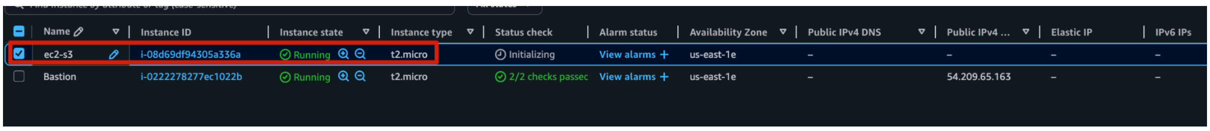




- **Network settings → Edit:**
  - **VPC:** `my-VPC`
  - **Subnet:** `private-1`
  - **Auto-assign public IP:** `Disable`
  - **Firewall (SG): Select security group**
    - **Security group name:** `ec2-s3-sg`
    - **Description:** `ec2-s3-sg`
    - **Inbound rules:**
      - **Type**: `SSH`
      - **Protocol:** `TCP`
      - **Port range:** `22`
      - **Source** :`bastion-sg`



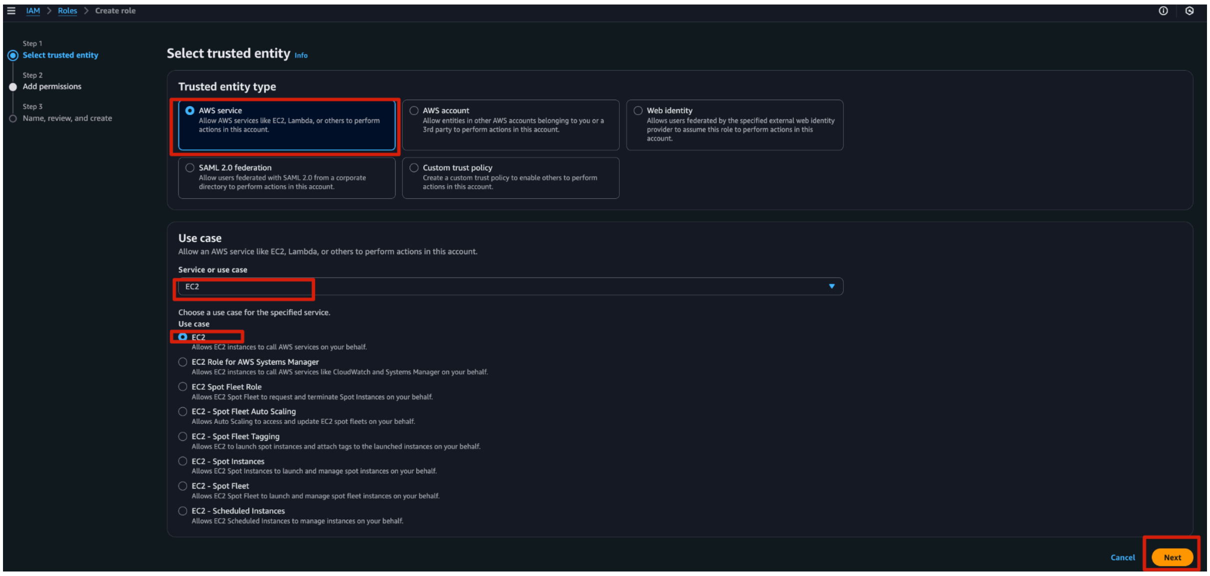
Launch instance butonuna bas



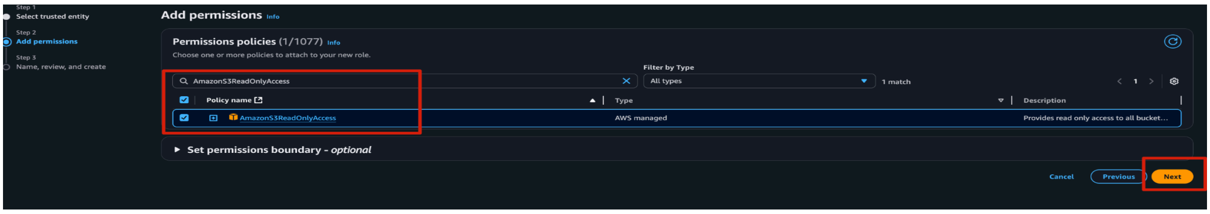
3. **IAM servisinde IAM Role oluştur**
    - IAM Role: S3 erişim yetkisi olan bir role ata (`AmazonS3ReadOnlyAccess`)

IAM→Role→Create Role

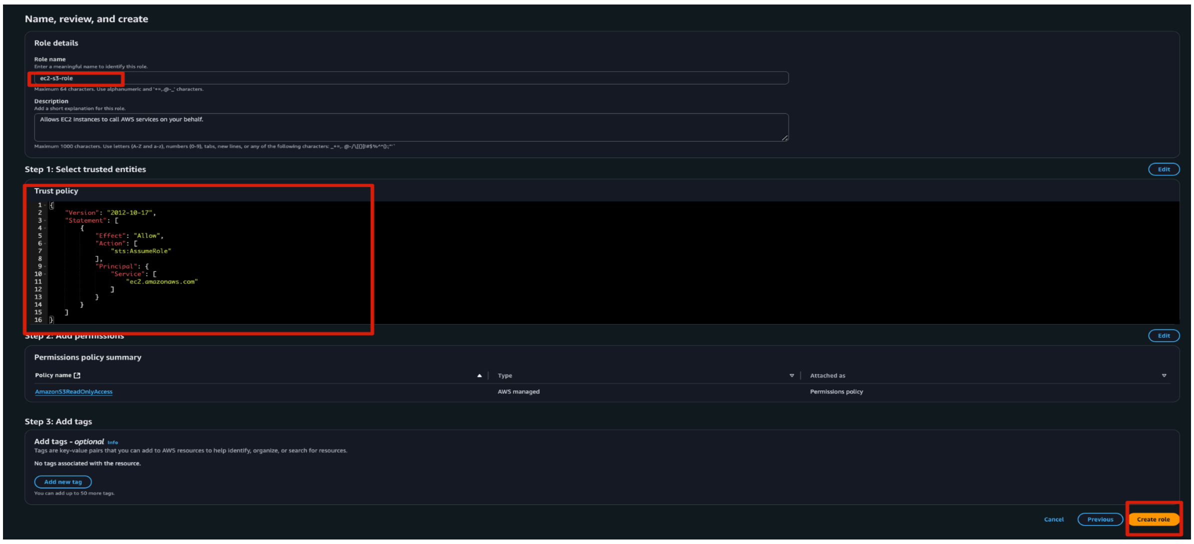
AWS Service seç→**EC2** seç ve **next** butonuna bas.



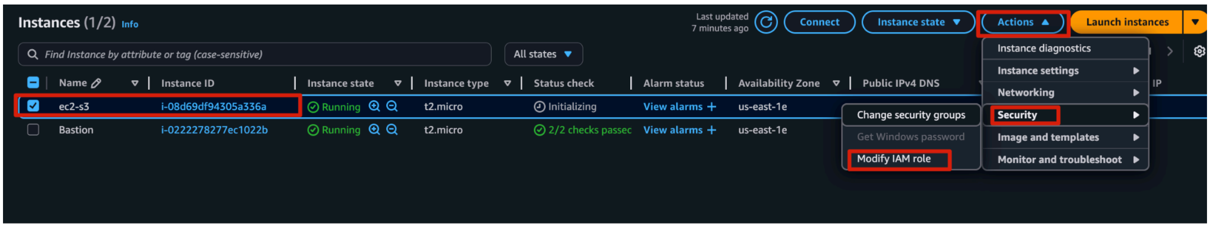
`AmazonS3ReadOnlyAccess` `permission seç ve` **`next`** `butonuna bas`



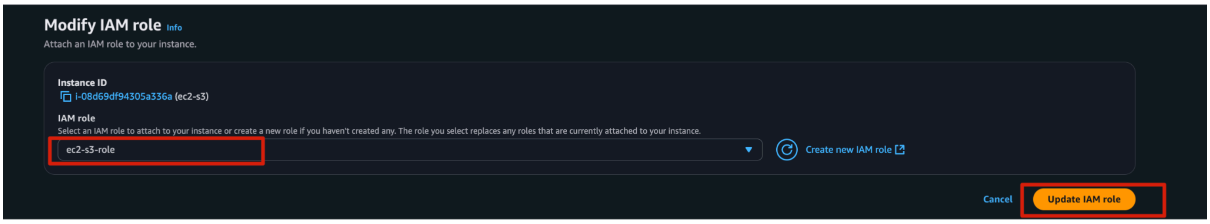
`Diğer ayarları default bırak veve` **`create role`** `butonuna bas`



4. **EC2 suncuya Role ata**
    - **Actions→Security→Modify IAM** role



    - **Ec2-s3-role** rseç Update IAM role butonuna bas

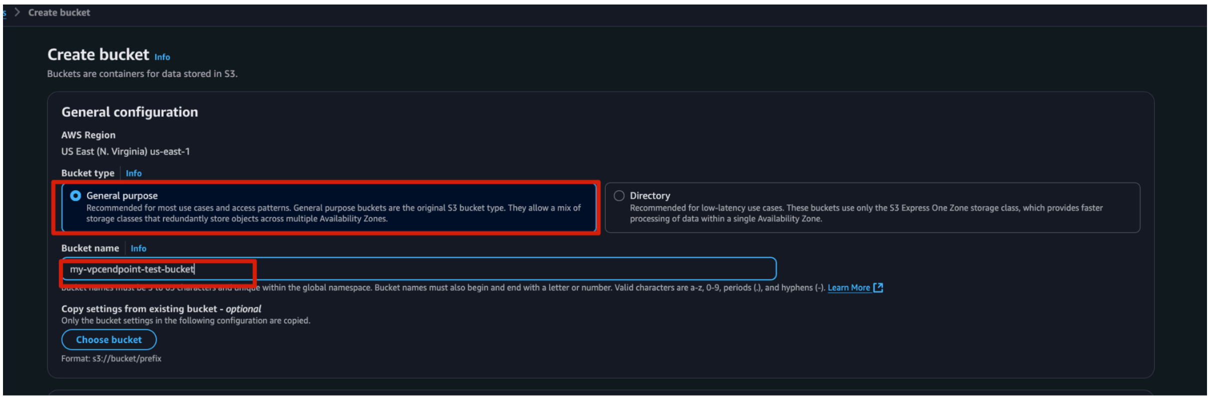


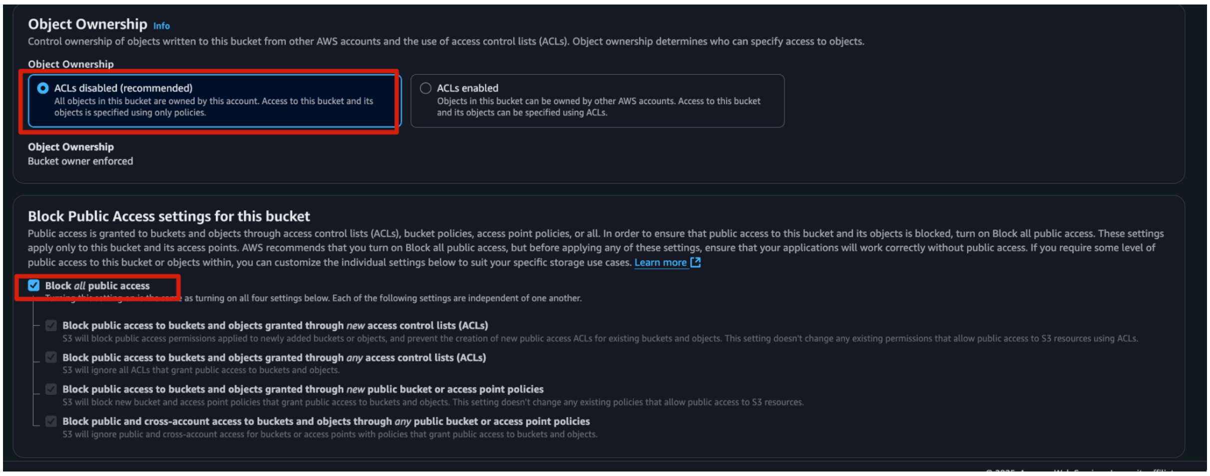
3️⃣ S3 Bucket Hazırlığı

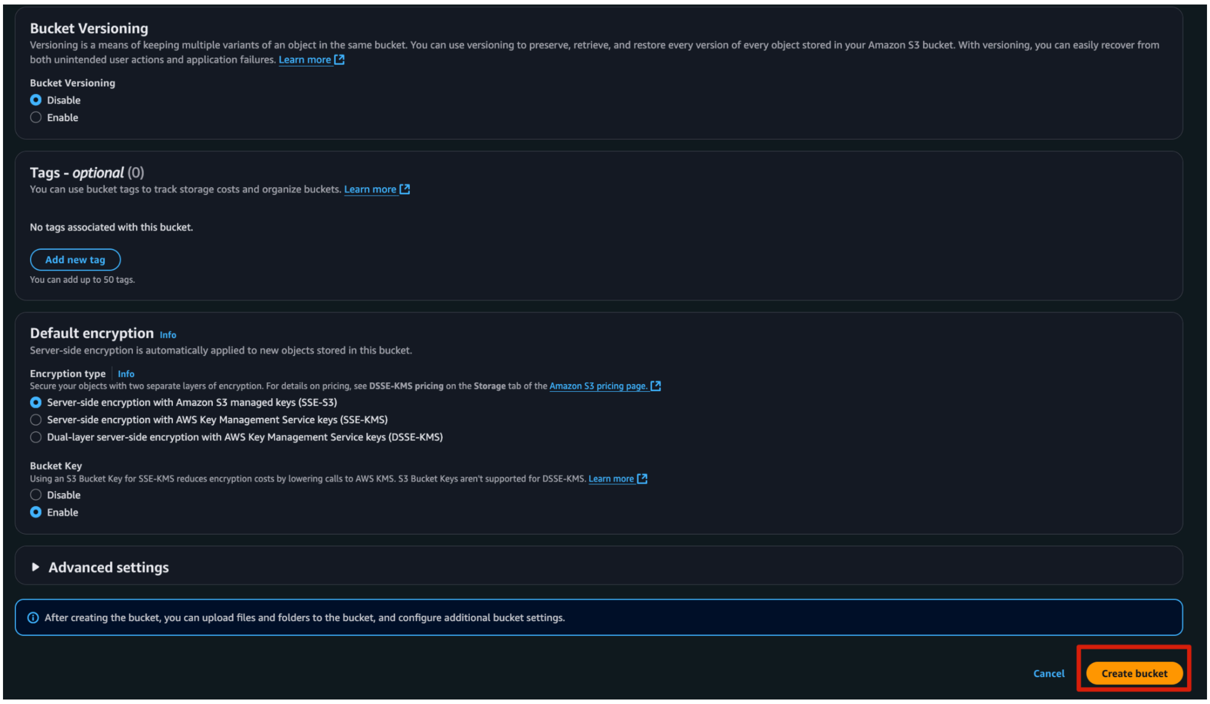
1. **S3 Bucket Oluşturma**

Amazon S3→Create Bucket

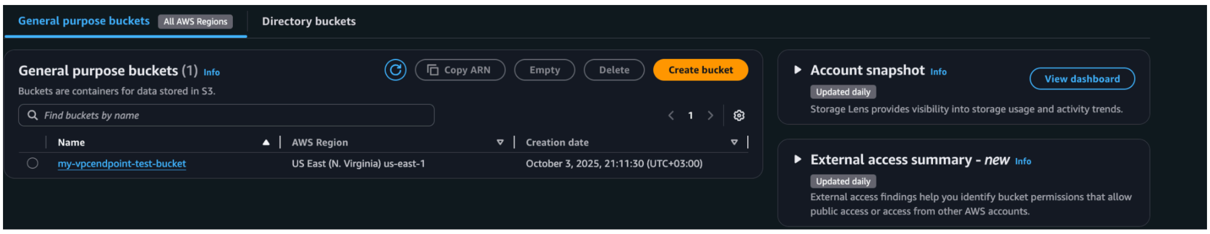
`Name: my-vpcendpoint-test-bucket`





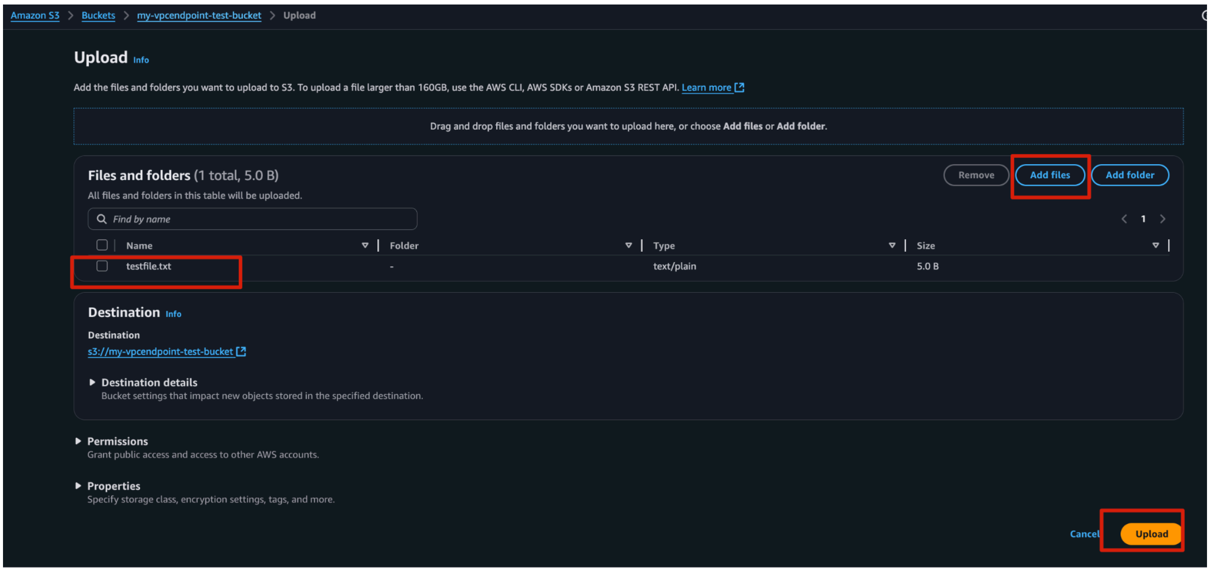


S3 bucket ismi verdikten sonra herşeyi default ayarlarında bırak ve **Create bucket** butonuna basarak s3 bucket oluştur.



2. **S3’e Dosya Yükleme**

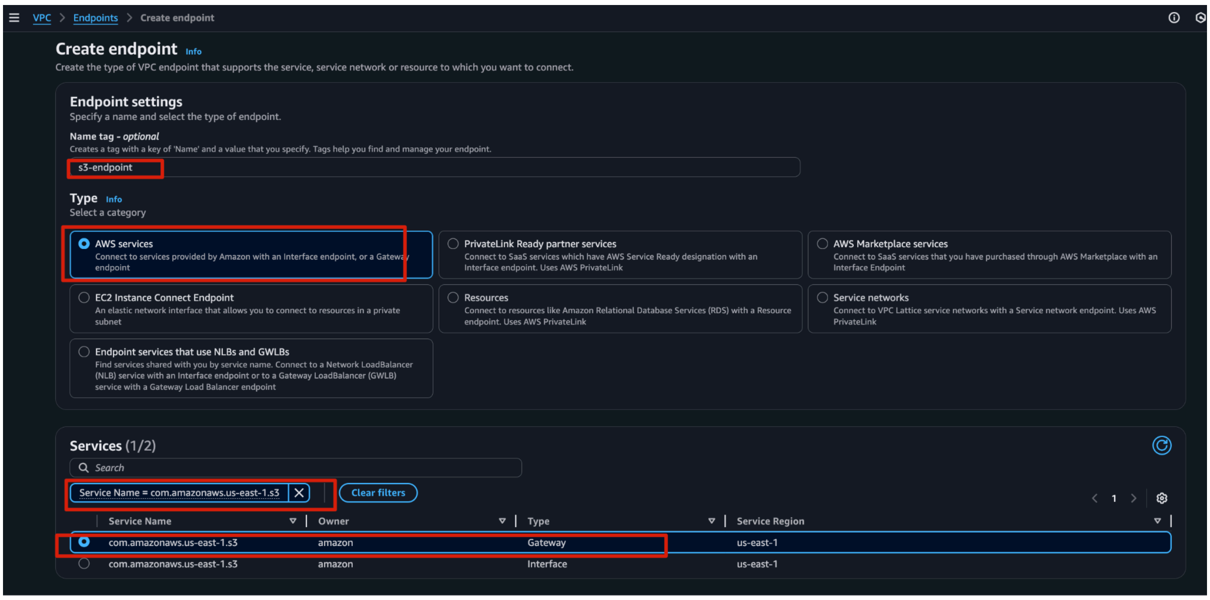
`(my-vpcendpoint-test-bucket)` Bucket’ı seçerek upload butonuna tıkla ve s3 bucket içine testfile.txt adında dosyayı ekle.



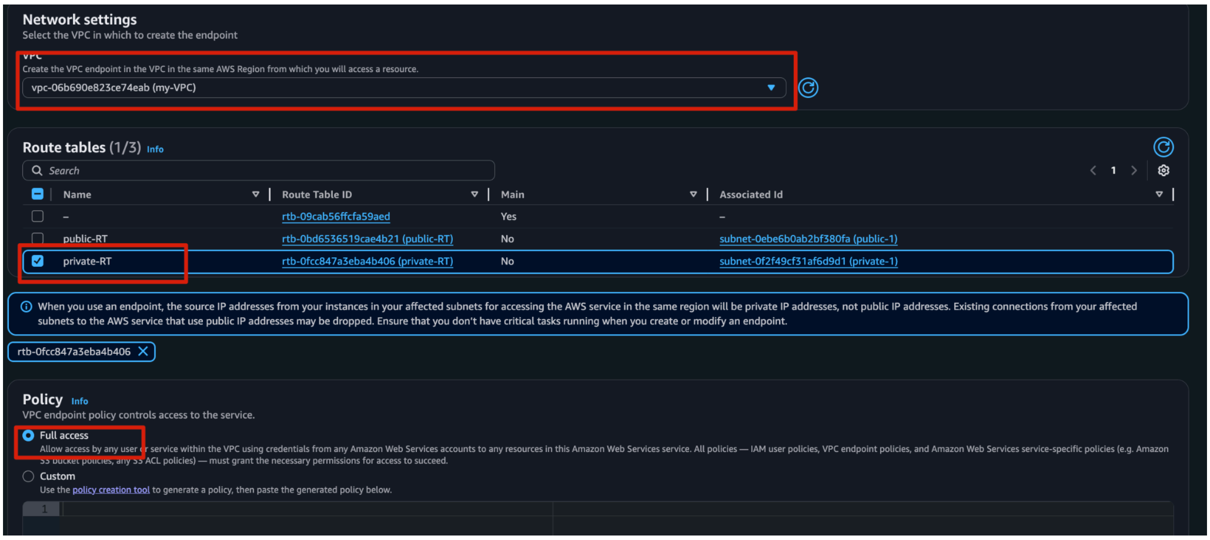
4️⃣ VPC Endpoint (Gateway) Kurulumu

1. VPC→**PrivateLink and Lattice**→ **Endpoints** → **Create Endpoint**

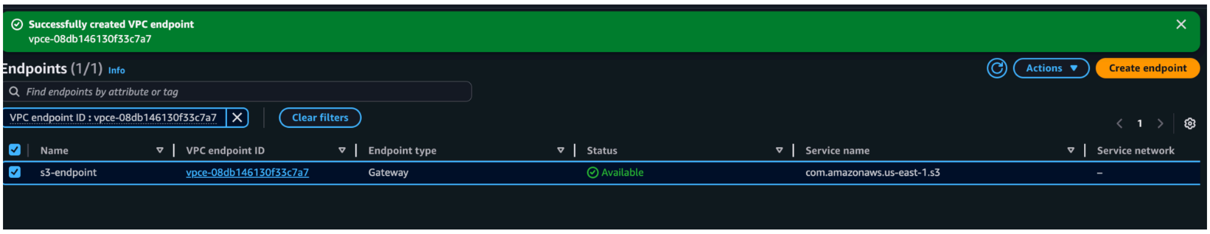
   - **Name:** s3-endpoint
   - **Service category**: AWS services
   - **Service Name**: `com.amazonaws.us-east-1.s3`
   - **Type**: Gateway



- **VPC**: `my-VPC`
- **Route Table**: Private Subnet'in Route Table'ını seç.
- Policy: Full Access (test için).

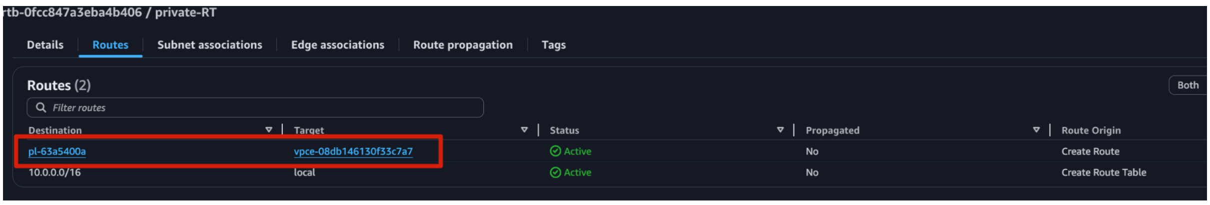


Create endpoint butonuna tıkla



📌 Bu işlem sonunda Route Table'a otomatik şu entry eklenir:

`pl-xxxx (Prefix List for S3)` → Target = VPC Endpoint



5️⃣ Test – EC2’den S3 Erişimi

1. Bastion Host üzerinden Private EC2’ye SSH ile bağlan.

```
ssh -i ec2_ssh_key.pem -o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W
%h:%p ec2-user@<Bastion_PublicIP>’ ec2-user@<ec2-s3-privateIP>
```

```
chmod 400 ec2_ssh_key.pem
ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p ec2-user@54.209.65.163' \
  ec2-user@10.0.2.14
```

```
(base) Downloads 👉 chmod 400 ec2_ssh_key.pem
(base) Downloads 👉 ssh -i ec2_ssh_key.pem \
-o 'ProxyCommand=ssh -i ec2_ssh_key.pem -W %h:%p ec2-user@54.209.65.163' \
  ec2-user@10.0.2.14
The authenticity of host '54.209.65.163 (54.209.65.163)' can't be established.
ED25519 key fingerprint is SHA256:zmbq86Xg5lKYXQqytinRh2o7zcOqqKVMeJQ9oSV2SKo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.209.65.163' (ED25519) to the list of known hosts.
The authenticity of host '10.0.2.14 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:APzYr96jJRpankk574qNC+qvzbTOdrnARpuz2nxSaPg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.14' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-2-14 ~]$
```

AWS CLI ile test et:

```
aws s3 ls s3://my-vpcendpoint-test-bucket
aws s3 cp s3://my-vpcendpoint-test-bucket/testfile.txt .
```

```
[ec2-user@ip-10-0-2-14 ~]$  aws s3 ls s3://my-vpcendpoint-test-bucket
2025-10-03 19:15:20          5 testfile.txt
[ec2-user@ip-10-0-2-14 ~]$ aws s3 cp s3://my-vpcendpoint-test-bucket/testfile.txt .
download: s3://my-vpcendpoint-test-bucket/testfile.txt to ./testfile.txt
[ec2-user@ip-10-0-2-14 ~]$ ls -l
total 4
-rw-r--r--. 1 ec2-user ec2-user 5 Oct  3 19:15 testfile.txt
[ec2-user@ip-10-0-2-14 ~]$
```

2. Eğer internet yok ama dosya iniyorsa → VPC Endpoint başarılı 🎉

```
[ec2-user@ip-10-0-2-14 ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
```

💡 Notlar

- **IAM Role olmadan** EC2, S3'e erişemez → mutlaka Role/Policy ekle.
- Bu senaryoda **NAT Gateway'e ihtiyaç yok** → maliyet düşer.